

**CAMBRIDGE**  
INTERNATIONAL EXAMINATIONS

**JUNE 2002**

**GCE Advanced Subsidiary Level  
Advanced International Certificate of Education**

**MARK SCHEME**

**MAXIMUM MARK : 50**

**SYLLABUS/COMPONENT :9709 /6, 0390 /6**

**MATHEMATICS  
(Probability and Statistics 1)**



UNIVERSITY of CAMBRIDGE  
Local Examinations Syndicate

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|                                                                                                                                                                                                                          |                                                    |                                                                                                                                                                                                                                                                                             |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 (i) not independent<br>$P(A) \times P(B) \neq P(A \text{ and } B)$                                                                                                                                                     | B1<br>B1dep 2                                      |                                                                                                                                                                                                                                                                                             |
| (ii) not mutually exclusive<br>$P(A \text{ and } B) \neq 0$                                                                                                                                                              | B1<br>B1 2                                         | Can be stated in words                                                                                                                                                                                                                                                                      |
| 2 both axes correct<br><br>points<br><br>median<br>IQ range                                                                                                                                                              | B1<br><br>M1<br>A1<br><br>B1ft<br>M1<br><br>A1ft 6 | For correct scales and labels on at least one axis<br><br>For points at upper bounds or 15.5 or 14.5<br>All correct and smooth curve or straight lines<br><br>On mid-points or upper bounds<br>For evaluating their UQ – theirLQ<br><br>For correct answer, ft on correct upper bounds only |
| 3 (i) $a$ 1 4 9 16<br>$P(A=a)$ $\frac{1}{2}$ $\frac{1}{6}$ $\frac{1}{6}$ $\frac{1}{6}$                                                                                                                                   | M1<br>A1<br><br>A1 3                               | For $A = 1, 4, 9, 16$ , or $1, 1, 1, 4, 9, 16$<br>Any three correct probabilities for 3 different vals of A<br>All correct                                                                                                                                                                  |
| (ii) $E(A) = 1 \times \frac{1}{2} + 4 \times \frac{1}{6} + 9 \times \frac{1}{6} + 16 \times \frac{1}{6}$<br>$= 5.33$<br>$\text{Var}(A) = 1^2 \times \frac{1}{2} + 4^2 \times \frac{1}{6} + \dots - (5.33)^2$<br>$= 30.9$ | M1<br>A1<br>M1<br>A1 4                             | For calculation of $\sum xp$ where $\sum p$ must be 1<br>For correct answer<br>For calculation of $\sum x^2 p - (\text{their } E(A))^2 \sum p$ need not be 1<br>For correct answer                                                                                                          |
| 4 (i) $-47.2/30 = -1.573$<br>OR $\sum x - \sum 110 = -47.2$ and $\sum 110 = 3300$<br><br>$\bar{x} = 110 - 1.573 = 108.4$<br><br>standard deviation = $\sqrt{\frac{5460}{30} - (-1.573)^2}$<br>$= 13.4$                   | B1<br><br>B1<br>M1<br>A1 4                         | For correct answer<br>For $\frac{5460}{30} - (\text{their coded mean})^2$<br>For correct answer                                                                                                                                                                                             |
| (ii) $z = \frac{110 - 107.6}{13.8} = 0.174$<br>$P(X > 110) = 1 - \Phi(0.174)$<br>$= 1 - 0.5691$<br>$= 0.431$                                                                                                             | M1<br>M1<br>A1 3                                   | For standardising, can have $\sqrt{13.8}$ on denom not $13.8^2$<br>For using tables correctly and finding a correct area from their $z$ .<br>For correct answer                                                                                                                             |

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|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5 (i) $\frac{7!}{2!} = 2520$                                                                                                                                                                                                                                                                 | M1<br>A1 2                                         | For dividing by 2 or 2!<br>For correct answer                                                                                                                                                                                                                                                                                                          |
| (ii) $\frac{5!}{2!} \times 3! = 360$                                                                                                                                                                                                                                                         | B1<br>M1<br>A1 3                                   | For 5! or equivalent<br>For multiplying by 3! or dividing by 2! or both<br>For correct answer                                                                                                                                                                                                                                                          |
| (iii) $4/7$ of 2520 = 1440<br><br>OR $6! + \frac{6!}{2!} + \frac{6!}{2!} = 1440$                                                                                                                                                                                                             | M2<br>A1<br><br>M1<br>A1<br>A1 3                   | For 4/7 of their (i)<br>For correct answer<br><br>For summing options for ending in 2, 6, 8<br>For correct options<br>For correct answer                                                                                                                                                                                                               |
| (ii) (i) $\mu = 3.6$<br><br>$\frac{2.8 - \text{their } \mu}{\sigma} = -0.4$<br><br>$\sigma = 2$                                                                                                                                                                                              | B1<br><br>M1<br>M1<br>A1 4                         | Stated or can be calculated later on<br><br>For equation relating $\mu$ or 3.6 and $\sigma$ . Must be standardised, can have $\pm 0.4$<br>Solving the correct equation or with a second correct equation relating $\mu$ and $\sigma$<br>For correct answer                                                                                             |
| (ii) $(0.6554)^2 \times (0.3446)^2 \times {}_4C_2$<br>$+ (0.6554)^3 \times (0.3446)^1 \times {}_4C_3 + (0.6554)^4$<br><br>$= 0.879$<br>$(= 0.3061 + 0.3881 + 0.1845)$<br><br>OR $1 - (0.3446)^4 - (0.6554)^1 \times (0.3446)^3 \times {}_4C_3$<br><br>$(= 1 - 0.0141 - 0.1072)$<br>$= 0.879$ | M1<br>B1<br>A1<br>A1<br><br>M1<br>B1<br>A1<br>A1 4 | For attempted binomial calculation of any 2 or 3 of P(2), P(3), P(4), needs 0.6554 in<br>For correct numerical expression for P(2) or P(3)<br>All in correct form<br>For correct answer<br><br>For calculation of 1 – any 2 or 3 of P(0), P(1), P(2)<br>For correct numerical expression for P(1) or P(2)<br>All in correct form<br>For correct answer |
| 7 (i) (a) $np = 11$<br>$np(1 - p) = 4.95$<br>$n = 20$ ( $p = 0.55$ )                                                                                                                                                                                                                         | B1<br>B1<br>M1<br>A1 4                             | For solving, need to find a value for $n$<br>For correct answer                                                                                                                                                                                                                                                                                        |
| (b) $P(X = 12) = (0.55)^{12} \times (0.45)^8 \times {}_{20}C_{12}$<br>$= 0.162$                                                                                                                                                                                                              | M1<br>A1 2                                         | For $(\text{their } p)^{12} \times (\text{their } q)^{n-12} \times k \neq 1$<br>For correct answer                                                                                                                                                                                                                                                     |
| (ii) $\mu = 100 \times 0.3 = 30$ , $\sigma^2 = 100 \times 0.3 \times 0.7$<br>$P(X < 35) = \Phi\left(\frac{34.5 - 30}{\sqrt{21}}\right)$<br>$= \Phi(0.9820)$<br>$= 0.837$ (exact)                                                                                                             | B1<br>M1<br><br>M1<br>A1 4                         | For both mean and variance correct, allow $\sigma = 21$<br>For standardising with or without cc, allow their 21 or their $\sqrt{21}$ in denom<br>For use of any continuity correction 34.5 or 35.5<br>For correct answer                                                                                                                               |